

Syed Shayan Mazahir

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Geomatics

Expected June 2030

EXPERIENCE

Robotics Team Lead & Engineer

2024 – 2025

Louise Arbour Secondary School

- Led 15-student team to First Place in RobOlympics competition
- Designed and built 4 autonomous robots including line-following and soccer robots
- Engineered voltage regulation circuits (5V to 15V), electrical schematics, and wiring harnesses
- Created CAD models for custom mechanical assemblies
- Implemented A* pathfinding and PID control algorithms in C++
- Integrated ultrasonic/IR sensors, gyroscopes, and motor encoders for real-time positioning
- Developed sensor fusion techniques for obstacle avoidance and human tracking
- Built modular platforms with interchangeable hardware (servo radar, LCD displays, H-bridge motor drivers)
- Trained 15 members in embedded C++ programming and circuit debugging with oscilloscopes/multimeters
- Created technical documentation with annotated schematics and code repositories

Programming Hub Mentor

2024 – 2025

Louise Arbour Secondary School

- Developed Flask web application with Google API integration serving 500+ students
- Improved student engagement by 60% through UX research and iterative testing
- Mentored 20+ students in Python programming, algorithms, and data structures
- Tracked learning outcomes to refine curriculum and teaching approaches
- Implemented SSL/HTTPS encryption and OAuth 2.0 authentication
- Ensured compliance with school privacy policies and security regulations

Community Operations & Infrastructure Lead

2022 – 2025

Aspect Systems

- Reduced incident response time from hours to minutes with automated Python workflow systems
- Implemented event-driven architecture with real-time alerting for scalable operations
- Contributed to C++-based anti-cheat system analyzing player behavior patterns
- Used statistical methods to detect exploits while minimizing false-positive rates
- Established documentation framework with Git version control
- Streamlined onboarding process for distributed contributors across multiple time zones

PROJECTS

Mr. Beep-A-Lot - Multi-Mode Autonomous Robot

2024 – 2025

- Built autonomous robot with 3 operational modes: obstacle avoidance, environment scanning, and human following
- Implemented IR remote control, dual ultrasonic sensors, servo-controlled radar, and I2C LCD display
- Developed Processing (Java) visualization for real-time radar data display via Serial communication
- Programmed personality-driven LCD messages and autonomous navigation with sensor fusion

games.random - AI-Powered Game Development Platform

2024 – Present

- Engineered few-shot learning prompt templates for Claude API
- Achieved 85%+ syntactically correct code generation from natural language descriptions
- Enabled non-programmers to create functional JavaScript games
- Built real-time validation pipeline with Monaco editor
- Reduced runtime errors by 40% through instant syntax highlighting and error feedback
- Open-sourced on GitHub with comprehensive documentation of LLM prompt engineering techniques

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, Java, Racket

Hardware & Embedded: Arduino, Raspberry Pi, VEX Platform, Ultrasonic/IR/Gyroscopic Sensors, Motor Controllers, PWM, PID Controllers, Circuit Design

Frameworks & Tools: Node.js, Express, Flask, Django, Git/GitHub, MongoDB, PostgreSQL, Processing, REST APIs